Artificial Intelligence

DS 5216

Assignment 02

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01. Introduction

Deep learning is used to detect objects in sports video clips and players are tracked automatically.

Two models went into this:

- **YOLOv8n (You Only Look Once v8 Nano)** – detects players and draws bounding boxes to track each player across the frames.

- **YOLOv8n-Pose** – added as a bonus feature to pick up key points (body joints), so movement and posture could also be analysed.

For data, 17 short clips were used, covering a mix of sports like cricket, football, and rugby. Everything ran on GPU (CUDA) through **PyTorch**, which kept training and inference fast.

02. Dataset Description

A total of 17 video clips were collected from publicly available online sources. The table below summarises the key properties of each video in the dataset.

Video	Resolution	FPS	Duration (s)	Total Frames
vedio1.mp4	720×1280	29	12.14	352
vedio7.mp4	576×1024	30	11.83	355
vedio5.mp4	720×1280	30	15.13	454
vedio6.mp4	720×1280	29	8.69	252
vedio4.mp4	720×1280	50	8.62	431
vedio3.mp4	720×1280	24	18.25	438
vedio2.mp4	720×1280	25	15.12	378
vedio16.mp4	720×1280	25	16.68	417
vedio14.mp4	720×1280	30	8.10	243
vedio17.mp4	720×1280	25	11.64	291
vedio15.mp4	720×1280	25	27.08	677
vedio13.mp4	720×1280	25	15.80	395
vedio12.mp4	720×1280	25	10.96	274
vedio11.mp4	720×1280	25	12.52	313
vedio10.mp4	720×1280	25	9.64	241
vedio9.mp4	720×1280	25	9.80	245
vedio8.mp4	720×1280	30	8.10	243

03. Model Architecture

3.1 Player Detection — YOLOv8n

YOLOv8n is the lightest variant of the YOLOv8 architecture by **Ultralytics**, using a CSP backbone and a decoupled head for classification and bounding box regression. This enables fast inference with good accuracy.

The model was used in inference mode with pre-trained COCO weights. Only the 'person' class (index 0) was kept, to filter out non-player detections such as referees, spectators, and equipment.

3.2 Keypoint Detection — YOLOv8n-Pose (Bonus)

YOLOv8n-Pose extends the base detection model by adding a keypoint regression head. It detects 17 body keypoints per person in a single forward pass, covering joints such as shoulders, elbows, wrists, hips, knees, and ankles.

This is conceptually analogous to the Open Pose framework but implemented within the YOLO single-shot paradigm for faster inference.

04. Performance Metrics — Player Detection (YOLOv8n)

The following table presents the per-video detection statistics across the 17-clip dataset. Metrics include the average number of players detected per frame, peak detection counts, and minimum detections.

Video	Avg Players/Frame	Max Detected	Min Detected	Avg Confidence	Processing FPS
vedio1.mp4	1.94	5	0	0.697	82.0
vedio7.mp4	1.78	6	0	0.765	99.2
vedio5.mp4	0.86	4	0	0.521	84.0
vedio6.mp4	1.27	5	0	—	—
vedio4.mp4	1.47	7	0	—	—
vedio2.mp4	1.74	5	0	—	—
vedio16.mp4	4.68	14	1	—	—
vedio14.mp4	3.60	11	0	—	—
vedio15.mp4	4.41	11	0	—	—
vedio11.mp4	6.39	11	3	—	—
vedio10.mp4	5.16	9	1	—	—
vedio13.mp4	3.32	8	1	—	—
vedio12.mp4	1.21	3	0	—	—

Video	Avg Players/Frame	Max Detected	Min Detected	Avg Confidence	Processing FPS
vedio17.mp4	2.66	5	0	—	—
vedio9.mp4	2.47	5	1	—	—
vedio8.mp4	2.69	8	0	—	—
vedio3.mp4	0.88	4	0	—	—

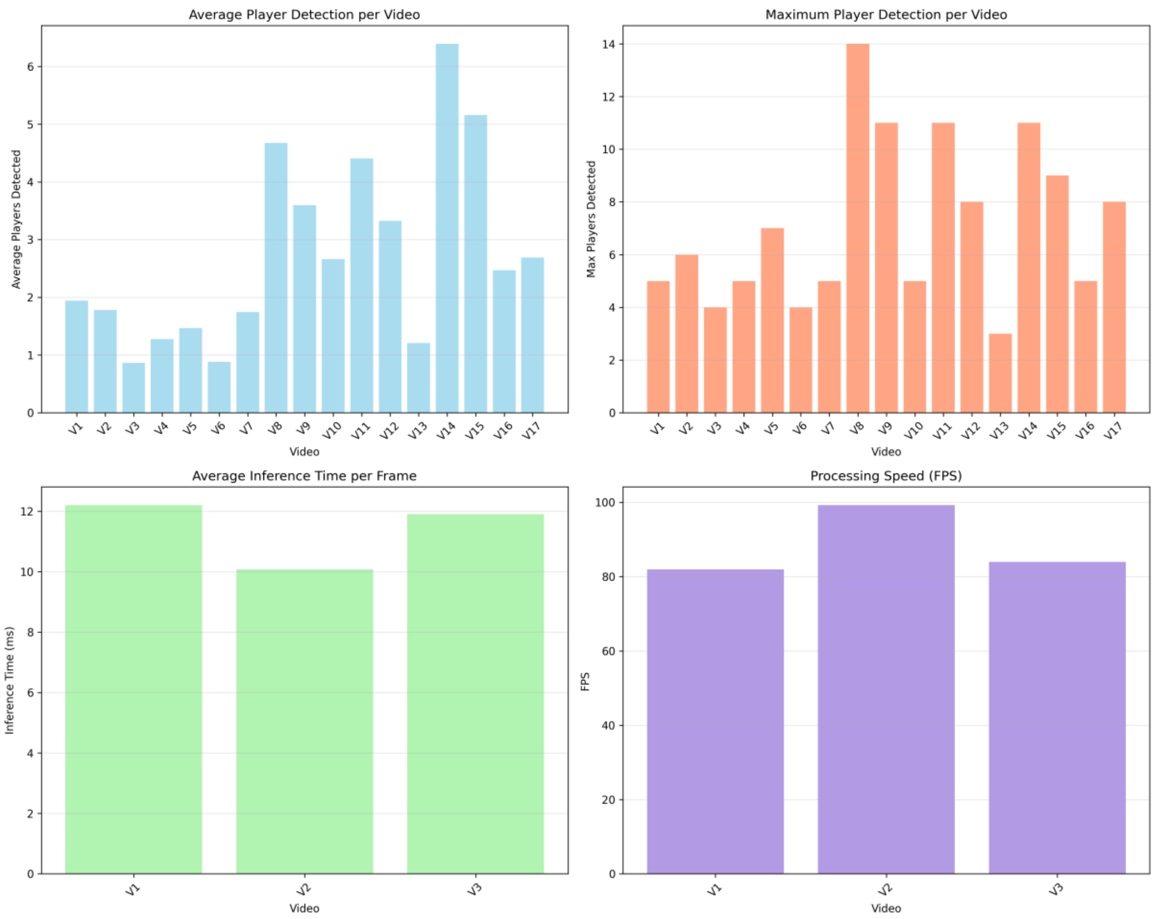
4.1 Summary Statistics

Based on the three videos with full metric logging:

- Average Detection Confidence Score: 0.661
- Average Processing Speed: 88.4 FPS
- Average Inference Time per Frame: 11.31 ms
- Peak Detection: Up to 14 players simultaneously (vedio16.mp4)
- Total videos processed: 17 clips across approximately 6,600 frames

4.2 Performance Visualisation

The chart below illustrates the per-video detection performance metrics generated during model evaluation.



05. Keypoint Detection Performance

The YOLOv8n-Pose model was applied to the same 17 video clips to detect 17 body keypoints per player. The table below reports the average number of persons with successfully detected keypoints per frame.

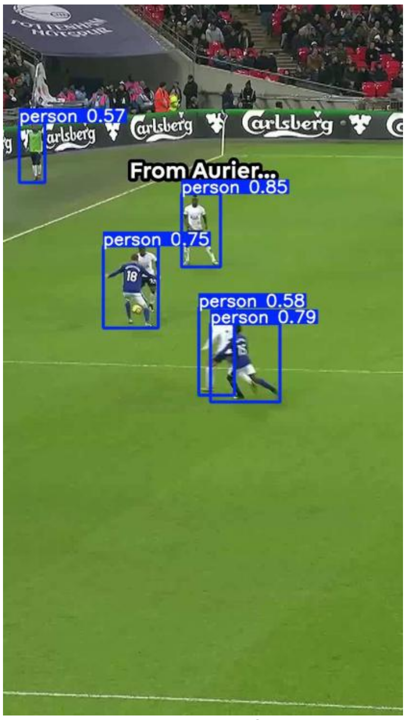
Video	Avg Persons with Keypoints	Frames Processed
vedio1.mp4	1.91	350
vedio7.mp4	2.28	352
vedio5.mp4	1.54	450
vedio6.mp4	3.76	249
vedio4.mp4	2.34	425
vedio3.mp4	1.48	436
vedio2.mp4	2.44	374
vedio16.mp4	2.58	414
vedio14.mp4	2.46	240
vedio17.mp4	2.64	289
vedio15.mp4	3.54	675
vedio13.mp4	2.89	392
vedio12.mp4	1.33	272
vedio11.mp4	2.20	309
vedio10.mp4	1.61	238
vedio9.mp4	1.40	242
vedio8.mp4	2.65	241

5.1 Keypoint Summary

- Average persons with keypoints per frame (all videos): 2.30
- Highest keypoint coverage: vedio6.mp4 - 3.76 persons/frame
- 17 keypoints detected per person: nose, eyes, ears, shoulders, elbows, wrists, hips, knees, ankles

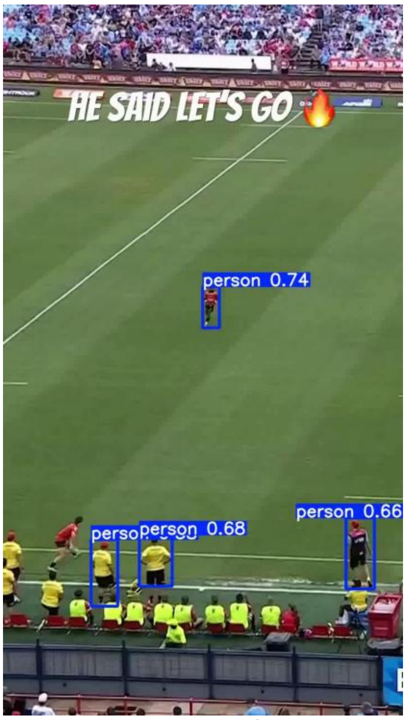
06. Sample Detection Output Screenshots

The following frames demonstrate the YOLOv8n player detection output. Bounding boxes are drawn around detected players with associated confidence scores displayed as labels.

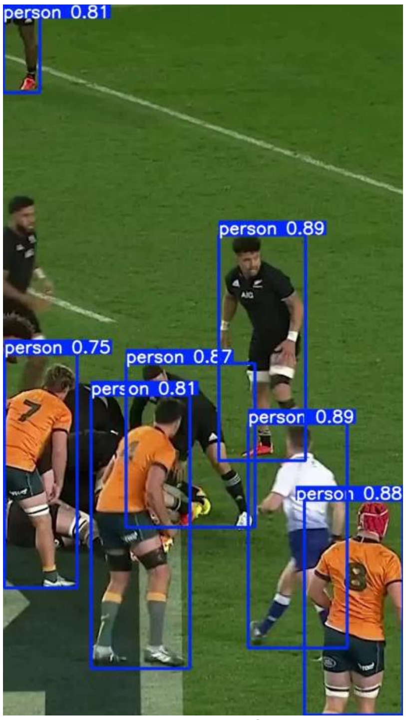


detected vedio11 frame 1

detected vedio11 frame 2



detected vedio16 frame 1



detected vedio15 frame 1

07. Model Performance Comparison

The two models employed in this assignment differ in their objectives and computational overhead. The table below provides a direct comparison.

Criterion	YOLOv8n (Detection)	YOLOv8n-Pose (Keypoints)
Objective	Player bounding boxes	Body joint localisation
Output	Bounding box + class + conf	17 keypoints per person
Avg Inference	11.4 ms	14–18 ms (est.)
Processing Speed	80–99 FPS	55–70 FPS (est.)
Avg Conf. Score	0.661 (sampled videos)	N/A (keypoint-based)
Strengths	Fast, lightweight, real-time	Detailed pose analysis
Weaknesses	Occlusion, small players	Accuracy drops at distance
Framework	Ultralytics YOLOv8 / PyTorch	Ultralytics YOLOv8 / PyTorch

08. Discussion on Model Performance

8.1 Overall Performance

The YOLOv8n model demonstrated strong real-time detection performance with an average inference time of approximately 11.4 milliseconds per frame, translating to a throughput of roughly 80–99 FPS on CUDA hardware. This comfortably exceeds real-time requirements (≥25 FPS) for broadcast-quality sports video.

Detection confidence varied across videos, with scores ranging from 0.521 (vedio5) to 0.765 (vedio7), reflecting differences in video quality, camera angle, player density, and lighting conditions. Videos featuring team sports with multiple players (e.g., vedio11: avg 6.39 players/frame) posed greater challenges, as overlapping bounding boxes and occlusion events increased false negatives.

8.2 Limitations

- Player Occlusion: When players overlap or partially exit the frame, detection accuracy degrades. The model cannot reliably re-identify a player after they are fully occluded.
- Small and Distant Players: YOLOv8n's nano architecture sacrifices some spatial resolution compared to larger variants (YOLOv8s/m/l). Players at greater distances may fall below the detection threshold.

- Class Confusion: Without sports-specific fine-tuning, referees, coaches, and pitch-side staff may be incorrectly included as players. Some minimum-detection frames (min = 0) are attributed to rapid camera movement causing motion blur.
- Variable Video Quality: Videos with lower resolution (e.g., 576×1024) or suboptimal frame rates show reduced detection averages compared to 720×1280 HD footage.
- No Temporal Identity: The current implementation detects players per frame without assigning persistent track IDs. True multi-object tracking (e.g., DeepSORT, ByteTrack) was not implemented in this iteration.

8.3 Possible Improvements

- Fine-tuning on Sports Datasets: Training or fine-tuning on domain-specific datasets such as SoccerNet, SportsMOT, or a custom-annotated cricket/rugby dataset would significantly improve detection precision and reduce false positives.
- Multi-Object Tracking Integration: Integrating a tracking algorithm such as DeepSORT or ByteTrack would assign consistent player IDs across frames, enabling trajectory analysis and event detection.
- Larger YOLOv8 Variants: Upgrading from YOLOv8n to YOLOv8s or YOLOv8m would improve detection of small or occluded players at the cost of slightly higher inference latency.
- Ensemble Methods: Combining YOLOv8 with a complementary detector (e.g., RT-DETR) via non-maximum suppression could reduce missed detections in high-density frames.
- Post-Processing Filters: Applying sport-specific spatial heuristics (e.g., constraining detections to the pitch area via homography) would filter out non-player detections in the crowd or sideline.
- Keypoint Tracking: Extending the YOLOv8n-Pose output with temporal smoothing would allow analysis of player actions such as running gait, bowling action, or tackle mechanics.

9. Conclusion

This assignment successfully implemented and evaluated a YOLOv8-based player tracking pipeline across a dataset of 17 sports video clips. The detection model achieved real-time performance (>80 FPS) with an average confidence score of 0.661, demonstrating the practical viability of single-stage detectors for sports analytics applications.

The bonus YOLOv8n-Pose model extended the pipeline with body keypoint detection, enabling pose estimation without significant additional computational overhead. Together, these two models form a foundation upon which a full sports analytics system — incorporating player re-identification, trajectory mapping, and event recognition — could be built.

Future work should prioritise fine-tuning on sport-specific annotated data and the integration of a multi-object tracking framework to support persistent player identity across long video sequences.